
# MD ABDULLAH IBNA MASUD

**AI/ML • Software • Data • IT & Creative**

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**Portfolio • GitHub • LinkedIn**


## Professional Summary


CSE graduate with hands-on experience in AI/ML, data analysis, and software development. Developed an undergraduate thesis project on low-resource Noakhali Bangla Speech Emotion Recognition using WavLM with parameter-efficient fine-tuning and controlled noise augmentation. Experienced in Python, data analysis, machine learning, and practical web-based application development. Seeking an entry-level opportunity in AI/ML, Data, or Software Engineering.


## Technical Skills

**AI/ML:** Machine Learning, Deep Learning, WavLM, Speech Emotion Recognition, PyTorch
**Python & Data:** Python, Pandas, NumPy, Matplotlib, Seaborn, Data Analysis
**Software & Web:** HTML, CSS, JavaScript, Node.js, REST APIs, Git, GitHub
**Database:** SQL, MySQL
**Productivity:** Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Google Colab, Kaggle, VS Code
**Additional / Fundamentals:** IT Support, Troubleshooting, Networking Fundamentals, MIS
**Creative:** Figma, Graphic Design, Logo Design

## Undergraduate Thesis -- Flagship Project

***Parameter-Efficient Fine-Tuning of WavLM for Low-Resource Noakhali Bangla Dialect Speech Emotion Recognition with Controlled Noise Augmentation*** *Mar 2026*

*Feni University*

- Developed a speaker-independent Speech Emotion Recognition system for low-resource Noakhali Bangla dialect using WavLM-Large.
- Built and evaluated the BNSER dataset containing 2,300 audio samples collected from 25 native Noakhali speakers across six emotion classes.
- Applied parameter-efficient fine-tuning by freezing the first 12 layers of the 24-layer WavLM encoder, combined with controlled noise augmentation and class-weighted cross-entropy loss.
- Achieved 88.70% accuracy, 0.8872 weighted F1, and 0.8881 balanced accuracy on the speaker-independent test set, with 48% fewer trainable parameters.
- Developed a web-based demo for practical speech emotion recognition using the trained model.

## Selected Project

**Speech Emotion Recognition – Interactive Web Demo** 2026

*Personal Project*

- Integrated the fine-tuned WavLM-based thesis model into a web-based interface where users can upload or record speech and receive predicted emotion categories.
- Demonstrates the practical deployment of the research model as an interactive AI application. Live Demo

## Education

**B.Sc. in Computer Science and Engineering** 2022 – 2025

Feni University CGPA: **3.59 / 4.00** (EQF Level 6) • Coursework: Data Structures & Algorithms, DBMS, Computer Networks, Software Engineering, Artificial Intelligence, OOP, Linear Algebra

## Courses & Certifications 2025

- **Machine Learning Foundations: A Case Study Approach** – Regression, Classification, Clustering, Recommender Systems, Deep Learning (97.61%)
- **Getting Started with Machine Learning Algorithms** – Linear & Logistic Regression, K-Means Clustering, Decision Tree, Random Forest, PCA, Q-Learning
- **Machine Learning using Python** – Supervised & Unsupervised Learning, Time-Series Modeling, Kernel SVM
- **AI/ML Projects** – ML Algorithms, DL Techniques, NLP, Model Evaluation & Optimization, Problem Solving

## Languages

Bangla – Native proficiency | English – Professional working proficiency